



Blended Learning System with Syllabus Mapping

A Capstone

Presented to Jeffred P. Lim

The Faculty

I-TECH COLLEGE BAGO CITY, INC.

Ramon Gonzaga Compound, 1084 Araneta St.

Brgy. Poblacion, Bago City

In Partial Fulfillment

Of the Requirements for the Degree

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

By:

Cristian Felix Demateo

Emmanuelle Margaux Nomat

Sherry Lou Belanio

Justine Joy Deniola

APPROVAL SHEET

AKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all the individuals who have contribute of this study. Without their valuable support this study would not have been possible.

First and foremost, we extend our deepest appreciation to our research instructor for his expertise and constant encouragement throughout the research process. His valuable insights and suggestions played a crucial role in shaping this study.

Furthermore, we would like to thank our classmates, friends, and family members for their unwavering support and understanding the research process. Their encouragement and belief in our abilities have been constant source of motivation.

Lastly, we express our gratitude to all authors, researchers, and scholars, whose work referenced to this study. Their contributions to the field have provided solid foundation for our research study.

ABSTRACT

The focus of this study is the development and evaluation of a Blended Learning System with Syllabus Mapping in I-Tech College Inc. Bago City, Negros Occidental. The system was designed to address issues in curriculum alignment, inconsistent instructional delivery, and gaps between learning outcomes, teaching strategies, and assessment methods. Existing challenges in blended learning implementation, such as unclear alignment of course objectives and duplication of content across online and face-to-face modalities, prompted the development of a more coherent framework.

This research employed a descriptive-developmental design and utilized surveys, interviews, observations, and document analysis involving faculty members and students. The findings from the descriptive phase served as the basis for developing an improved syllabus mapping framework aligned with blended learning delivery. The Agile Model was adopted to guide the developmental process, allowing continuous refinement based on feedback and identified needs.

Following the implementation of the proposed framework, improvements were observed in curriculum coherence, clarity of learning outcomes, instructional organization, and assessment alignment. The structured syllabus mapping contributed to enhanced instructional quality and increased student engagement within the blended learning environment. The study concludes that integrating a systematic syllabus mapping approach strengthens the effectiveness of blended learning and may serve as a reference model for institutions seeking to improve curriculum alignment and instructional delivery.

TABLE OF CONTENTS

	Pages
TITLE PAGE	i
APPROVAL SHEET	ii
ACKNOWLEDGEMENT	iii
ARSTRACT	iv
TABLE OF CONTENTS	v
LIST OF TABLES	vii
LIST OF FIGURES	ix
Chapter 1 - INTRODUCTION	
Background of the Study	1
Objectives of the Study	2

Conceptual Framework of the Study	3
Significance of the Study	4
Scope and Limitations	5
Definition of Terms	6
Chapter II - RELATED LITERATURE AND PRIOR ARTS RESEARCH	
Related Literatures	10
Prior Arts	11
Synthesis	12
Chapter III - METHODOLOGY	
System Design	32
Research Design	35
Context Flow Diagram	39
Data Flow Diagram	40
Entity-Relationship Diagram	41
Application Architecture	42
Use Case Diagram	44
User Interface Design	45
Software Requirements	46
Hardware and Other Required Devices	47
Chapter IV - PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA	
Chapter V - SUMMARY, CONCLUSION, AND RECOMMENDATIONS	
Summary of Findings	49

Conclusion	50
Recommendation	51
References	52

APPENDICES

A Certificate of Plagiarism	53
B Letter to the Client	54
C Evaluation Instrument for Experts	55
D Evaluation Instrument for Clients	56
E Curriculum Vitae	57

List of Tables

Table 1 The title of your first table

List of Figures

Figure 1 Conceptual Framework of the study

CHAPTER I

INTRODUCTION

Background of the Study

Hybrid learning has emerged as an influential approach in education, that combines the best of both traditional face-to-face and online activities. This approach supports variability in both teaching and learning styles. But in a blended environment that's not the end all and be all because it's important to make sure each lesson, activity, or assessment still aligns with the big picture: your course syllabus.

Many schools and universities teachers have the need to manage their course content (such as lesson plans, online materials or classroom activities), through various disparate tools/algorithms. This sometimes results in contradictions, omissions and inability to handle whether or not the content is covered completely. And students may have difficulty in seeing how online activities and their face-to-face equivalents help to satisfy course requirements.

The proposed blended learning system with syllabus mapping is to solve these issues by allowing the lesson, learning activity, and assessment to be centralized with the syllabus content as a reference. The system assists the instructor in tracking lesson progress, completed and incomplete progress and the relationship between teaching strategies and outcomes. It also gives students an overview of what they are going to learn, and how each activity fits within the whole course.

By implementing a blended learning system with syllabus mapping, educational institutions can improve lesson organization, transparency, and instructional effectiveness, leading to better learning experiences and academic outcomes.

Objectives of the Study

This study aims to develop a blended learning system with syllabus mapping that supports effective lesson tracking and alignment of learning activities.

Specifically, the study aims to:

1. Develop a system that maps lessons, activities, and assessments directly to syllabus objectives.
2. Provide instructors with tools to monitor lesson progress and syllabus coverage in a blended learning environment.
3. Enable students to track their learning activities and understand their progress in the course.
4. Improve lesson organization and reduce missed or duplicated topics in blended learning.
5. Support effective teaching and learning through a centralized and organized learning system.

Conceptual Framework of the study

INPUT

PROCESS

OUTPUT

OUTCOME

SDLC
(AGILE)

BLENDED LEARNING
SYSTEM WITH
SYLLABUS MAPPING

- Map lessons to syllabus objectives
- Track student progress in real-time
- Detect duplicate topics
- Monitor curriculum coverage
- Blended learning support
- Centralized learning management

FEEDBACK

Significance of the Study

This study is significant to the following groups:

- Students - The system helps students clearly understand their lessons, activities, and progress, making it easier for them to follow course requirements in a blended learning setup.
- Instructors - The system assists instructors in organizing lessons, tracking syllabus coverage, and monitoring student progress more efficiently.
- Administrator - The system enables administrators to oversee lesson delivery, monitor program implementation, and ensure consistency across blended learning courses.

Scope of the Study

This study focuses on the implementation of a Blended Learning System with Syllabus Mapping in the teaching and learning process. It examines how the combination of face-to-face classes and online learning, guided by syllabus mapping, helps organize lessons, deliver content effectively, and maintain learning continuity.

The study aims to evaluate the effectiveness of the blended learning system in aligning learning objectives, instructional activities, and assessments through syllabus mapping. It also looks into how syllabus mapping assists teachers in managing course content and how students perceive its usefulness in understanding lesson flow, requirements, and expectations.

The study is conducted in selected college-level educational institutions within I-Tech College Inc. Bago City. It is limited to this area and does not include institutions outside the chosen setting.

The research covers one academic semester, during which data are gathered and analyzed. The participants include teachers and students who regularly use the blended learning system in their classes.

In terms of methodology, the study uses a descriptive research design. Data are collected through surveys, interviews, and system evaluation tools to assess the system's usability, effectiveness, and the challenges encountered during its use.

Limitations of the Study

This study has several limitations that may affect the generalizability of the results. First, the number of participants is limited to a specific group of teachers and students, which may not represent other institutions or educational levels.

Second, the study is conducted over a short period of one semester; therefore, it does not measure the long-term effects of the blended learning system with syllabus mapping on students' academic performance.

Third, data collection may be affected by participants' availability and willingness to provide complete and honest responses, which could limit the depth of the findings.

Fourth, technological factors such as internet access, availability of devices, and users' familiarity with digital tools may influence how effectively the system is used.

Lastly, the study may be influenced by response bias, as the data are largely based on self-reported surveys and interviews. External factors, including institutional policies and unexpected changes in learning modalities, may also affect the results but are beyond the control of the researcher.

Statement of the Problem

Despite the increasing use of blended learning in educational institutions, many instructors and students experience difficulties in managing and tracking lessons effectively. Learning materials, activities, and assessments are often handled through different platforms, making it hard to ensure that all syllabus topics are properly covered and aligned with learning objectives.

Instructors face challenges in monitoring lesson progress and identifying which syllabus competencies have already been addressed or missed. At the same time, students may feel confused about their learning progress and the connection between online activities and face-to-face discussions.

This study aims to address these problems by developing a blended learning system with syllabus mapping that supports organized lesson tracking, improves alignment with syllabus objectives, and enhances both teaching and learning experiences.

Specifically, this study seeks to answer the following questions:

1. How can a blended learning system effectively map lessons to syllabus objectives?
2. How can the system help instructors track lesson progress and coverage?
3. How can the system help students monitor their learning activities and requirements?

Definitions of terms

Blended Learning System

Defining Blended Learning Conceptually: A combination of face-to-face instruction and online learning which offers greater flexibility and more effective learning opportunities.

Operational Definition: In this sense, it involves the classroom teaching also putting online lessons and activities through a Learning Management System (LMS) of the selected institution.

Syllabus Mapping

Conceptual Description: a curriculum planning process that clearly articulates how learning outcomes, teaching and learning experiences and assessment will be aligned and assessed in competing whole provide/context of a national examination.

Operational Definition: In this regard, it is the integration of course objectives, topics, activities and assessments in a seamless manner between the online and face-to-face components of the course. Curriculum Alignment Conceptual Description: Agreement or consistency between learning goals, instruction and assessment.

Operational Definition: In this study, it represents the consistency among course objectives, teaching learning activities and assessment in the blended system of education.

Learning Management System (LMS)

Concept: An online eLearning environment, where courses and other learning content can be presented, administrated and tracked.

Operational Definition: in this study it is considered as the online platform where institutions upload materials, test students and monitor progress.

Learning Outcomes

Conceptual Definition: Statements that define what students should know, be able to do, and understand as a result of completing a course.

Definition in Practice: In this study, it is used to mean specific curriculum objectives and the course content assessed through quizzes, projects, exams.

Student Engagement

Conceptual Definition: The degree with which students are actively and passively involved in learning.

Operational Definition: As used in this investigation, it is defined to be the participation of students in on-line discussion board activities, submitting assignments, attendance and involvement in classroom lecture based interactions.

Chapter II

RELATED LITERATURE AND PRIOR ARTS SEARCH

PRIOR ARTS

This section presents local and international systems, institutional implementations, and related technologies relevant to blended learning, syllabus mapping, and curriculum alignment. Each prior art is discussed in relation to its contribution, strengths, and identified limitations that justify the development of the proposed Blended Learning System with Syllabus Mapping.

A. Local Prior Arts (Philippines)

Moodle-Based LMS Implementations in Philippine State Universities

During the COVID-19 pandemic, many Philippine State Universities adopted Moodle as their primary Learning Management System to support flexible and blended learning delivery. Moodle, being an open-source platform, allows institutions to manage course materials, quizzes, assignments, and student progress (Moodle, 2023). Studies in Philippine higher education institutions revealed that Moodle significantly improved accessibility and communication between instructors and students during remote learning (Basilalaia & Kvavadze, 2020). However, despite its flexibility, most institutional deployments focused primarily on content delivery and assessment management. Structured syllabus mapping and automated alignment of lessons to course learning outcomes were often handled manually by instructors, creating potential inconsistencies in curriculum tracking.

University of the Philippines – University Virtual Learning Environment (UVLe)

The University of the Philippines developed UVLe as an institutional LMS to support online and blended instruction. UVLe provides centralized access to learning materials, discussion forums, and grade management tools. According to institutional reports, UVLe enhanced continuity of instruction and digital collaboration across campuses. However, while UVLe effectively manages course distribution and evaluation, it does not inherently integrate structured syllabus mapping tools that directly connect weekly lessons to defined program outcomes. Curriculum alignment remains dependent on instructor input

and external documentation, consistent with findings that many LMS platforms lack embedded constructive alignment systems (Biggs & Tang, 2011).

De La Salle University - AnimoSpace (Canvas LMS)

De La Salle University adopted Canvas through its AnimoSpace platform to facilitate blended and flexible learning. Canvas provides advanced analytics, outcome tracking, and modular content organization (Instructure, 2023). While it supports outcome-based education (OBE), structured syllabus mapping depends heavily on how instructors configure course modules. Research indicates that although LMS analytics improve transparency, curriculum alignment tools require systematic configuration and training to ensure consistent implementation (Sun et al., 2008). Therefore, automated lesson-to-syllabus linkage is not inherently standardized.

Polytechnic University of the Philippines - Open University System

The Polytechnic University of the Philippines implemented blended learning through digital modules and online platforms to support flexible education. While the system enabled continuity during pandemic disruptions, lesson progression and syllabus coverage were largely monitored through manual documentation. According to Graham (2006), effective blended learning requires structured instructional planning. Without integrated syllabus tracking tools, institutions may encounter duplication of content or incomplete coverage of learning outcomes.

Mapúa University Fully Online and Modular Model

Mapúa University pioneered a digital-first learning structure using modular and quarter-based systems. The university adopted digital platforms to ensure structured content sequencing. Although the modular structure supports organized delivery, automated lesson-to-outcome mapping within the system is not universally embedded. Research on modular instruction emphasizes that while modularization enhances clarity, integration with digital tracking systems is necessary for real-time curriculum monitoring (Horn & Staker, 2015).

CHED Flexible Learning Framework

The Commission on Higher Education (CHED) issued policies promoting flexible and blended learning models to ensure continuity of education. CHED emphasizes constructive alignment, outcome-based education, and instructional monitoring. However, while policy frameworks mandate alignment, they do not prescribe standardized digital tools for syllabus mapping. This creates variability in institutional implementation, reinforcing the need for integrated technological solutions.

Batangas State University Smart Campus Initiative

Batangas State University has implemented smart campus initiatives integrating digital infrastructure, including Learning Management Systems (LMS), online enrollment systems, academic dashboards, and student monitoring tools. These initiatives aim to modernize academic processes, improve data accessibility, and enhance institutional efficiency.

The LMS integration enables centralized content distribution, digital submissions, and performance tracking. Faculty members can upload course materials, record grades, and communicate with students through a unified system. From an administrative standpoint, academic monitoring tools allow departments to track enrollment trends, completion rates, and student progress.

However, while operational efficiency has improved, structured curriculum alignment features remain limited. Specifically, there is minimal system-level mapping between syllabus learning objectives, weekly lesson plans, instructional activities, and assessment tasks. Without embedded alignment tools, faculty members must manually ensure that learning outcomes are consistently addressed. This limitation reduces the system's ability to provide analytics on whether course delivery adheres to intended curriculum design principles, particularly under Outcome-Based Education (OBE) frameworks.

Technological University of the Philippines LMS Implementation

The Technological University of the Philippines adopted LMS platforms to support blended and flexible learning modalities. The system allows instructors to upload learning materials, administer quizzes, manage assignment submissions, and compute grades automatically. This transition significantly enhanced instructional continuity, particularly during remote learning periods.

Despite these advancements, syllabus monitoring remains largely external to the LMS environment. Faculty members often maintain separate course outlines, curriculum matrices, or OBE alignment documents outside the system. As a result, there is no automated mechanism that verifies whether uploaded materials, assessments, and activities correspond directly to specific learning outcomes.

This separation creates a documentation gap: while the LMS effectively supports instructional delivery, it does not inherently enforce constructive alignment. The lack of integrated curriculum-mapping dashboards limits institutional oversight in evaluating whether course implementation matches approved syllabus structures.

Local Capstone LMS Prototypes

In several Philippine higher education institutions, undergraduate Information Technology students have developed LMS prototypes as part of capstone requirements. These systems commonly feature user authentication, grade computation, attendance tracking, content uploads, messaging systems, and performance dashboards.

Technically, many prototypes demonstrate innovation in database design, interface usability, and system integration. Some incorporate analytics for student performance prediction or automated notifications for at-risk learners.

However, most capstone projects prioritize functional completeness over pedagogical structure. Curriculum alignment mechanisms—such as mapping course outcomes to specific lessons, linking assessments to measurable competencies, or generating outcome attainment reports—are rarely embedded in the system architecture.

This pattern reflects a broader trend in system development: emphasis is placed on administrative and operational capabilities rather than academic quality assurance. Consequently, while these prototypes improve efficiency, they do not systematically support outcome-based curriculum monitoring.

Outcome-Based Education (OBE) Implementation in Philippine HEIs

Philippine higher education institutions widely implement OBE frameworks in compliance with standards from the Commission on Higher Education (CHED). OBE emphasizes clearly defined learning outcomes, measurable competencies, and alignment between instruction and assessment. Institutions are required to demonstrate that graduates achieve program-level and course-level outcomes.

According to John Biggs and Catherine Tang (2011), constructive alignment ensures coherence among intended learning outcomes (ILOs), teaching and learning activities (TLAs), and assessment tasks (ATs). This model requires that every instructional activity and evaluation method directly supports specified outcomes.

Although OBE is structurally embedded in curriculum documents, course syllabi, and accreditation reports, its digital integration into LMS platforms remains inconsistent. Many institutions rely on manual encoding of outcome matrices, spreadsheets, or printed documentation for accreditation compliance. LMS platforms typically do not include automated outcome-tagging features, real-time alignment dashboards, or outcome attainment analytics.

This disconnect highlights a significant research and development opportunity: the need for LMS systems that integrate structured syllabus mapping, automated OBE tracking, and curriculum alignment visualization. Bridging this gap would strengthen institutional quality assurance, improve transparency, and enhance data-driven academic decision-making.

B. International Prior Arts

Google Classroom (Google LLC)

Google Classroom provides digital assignment management, communication tools, and grading systems. While widely adopted globally, it primarily supports content distribution rather than structured curriculum mapping. Research indicates that while Google Classroom improves efficiency, it lacks integrated outcome-alignment mechanisms (Al-Marooof & Al-Emran, 2018).

Canvas LMS (Instructure Inc.)

Canvas supports outcome-based tracking, analytics, and modular design. Institutions can manually align outcomes to assessments. However, automated syllabus mapping across weekly lessons requires

configuration. Studies show that LMS effectiveness depends on usability and alignment strategies (Sun et al., 2008).

Blackboard Learn

Blackboard offers comprehensive course management tools. It includes outcome assessment modules but requires institutional customization. Research emphasizes that LMS platforms must integrate constructive alignment to maximize instructional effectiveness (Garrison & Kanuka, 2004).

Moodle LMS (Global Implementation)

Moodle remains one of the most widely used open-source LMS platforms worldwide. It supports plugins for outcome mapping. However, default configurations do not automatically link syllabus components unless specifically structured.

Brightspace by D2L

Brightspace includes learning analytics and competency tracking tools. It supports outcome-based education models. However, similar to other LMS platforms, structured lesson-to-syllabus automation requires advanced setup.

Schoology (PowerSchool)

Schoology integrates instructional resources and assessments. While effective for blended learning environments, curriculum mapping tools are limited without external integration.

Open edX

Open edX supports modular learning and objective alignment in MOOC environments. However, it focuses primarily on large-scale online education rather than institutional blended tracking systems.

Atlas Curriculum Mapping

Atlas is a dedicated curriculum mapping software used internationally. It focuses on structured alignment of standards and outcomes. However,

it operates separately from LMS platforms, creating system fragmentation.

Coursera Platform

Coursera structures courses based on defined objectives and modules. While highly organized, it is not designed for institutional syllabus tracking or instructor-level alignment monitoring.

Constructive Alignment Framework (Biggs & Tang)

The constructive alignment model emphasizes that learning outcomes, teaching activities, and assessments must be interconnected. While widely adopted theoretically, integration into LMS architecture remains inconsistent across platforms.

Synthesis

The reviewed local and international systems demonstrate significant advancement in blended learning implementation and LMS functionality. However, a consistent gap remains in integrating structured syllabus mapping directly within learning management systems. Many platforms provide content management and assessment tools but rely on manual alignment processes to ensure curriculum coherence. This study addresses this gap by developing a Blended Learning System with built-in Syllabus Mapping that directly connects syllabus objectives, lessons, assessments, and student progress tracking in a centralized and automated platform.

References (APA 7th Edition Format)

Al-Marroof, R. A. S., & Al-Emran, M. (2018). Students' acceptance of Google Classroom: An exploratory study using PLS-SEM approach. *International Journal of Emerging Technologies in Learning*, 13(6), 112-123. <https://doi.org/10.3991/ijet.v13i06.8275>

Basilaia, G., & Kvavadze, D. (2020). Transition to online education in schools during a SARS-CoV-2 coronavirus (COVID-19) pandemic in Georgia. *Pedagogical Research*, 5(4), em0060. <https://doi.org/10.29333/pr/7937>

Biggs, J., & Tang, C. (2011). *Teaching for quality learning at university* (4th ed.). Open University Press.

Blackboard Inc. (2023). *Blackboard Learn: Product overview*. <https://www.blackboard.com>

Commission on Higher Education. (2020). *CHED memorandum order on flexible learning*. <https://ched.gov.ph>

Coursera Inc. (2023). *About Coursera*. <https://www.coursera.org>

De La Salle University. (2023). *AnimoSpace learning management system*. <https://www.dlsu.edu.ph>

D2L Corporation. (2023). *Brightspace learning platform*. <https://www.d2l.com>

Garrison, D. R., & Kanuka, H. (2004). Blended learning: Uncovering its transformative potential in higher education. *The Internet and Higher Education*, 7(2), 95-105. <https://doi.org/10.1016/j.iheduc.2004.02.001>

Graham, C. R. (2006). Blended learning systems: Definition, current trends, and future directions. In C. J. Bonk & C. R. Graham (Eds.), *The handbook of blended learning: Global perspectives, local designs* (pp. 3-21). Pfeiffer.

Google LLC. (2023). *Google Classroom overview*. <https://classroom.google.com>

Horn, M. B., & Staker, H. (2015). *Blended: Using disruptive innovation to improve schools*. Jossey-Bass.

Instructure. (2023). *Canvas learning management system*.
<https://www.instructure.com>

Mapúa University. (2023). *Fully online education model*.
<https://www.mapua.edu.ph>

Moodle Pty Ltd. (2023). *About Moodle*. <https://moodle.org>

Open edX. (2023). *Open edX platform overview*. <https://open.edx.org>

Polytechnic University of the Philippines. (2023). *Open University System*. <https://www.pup.edu.ph>

PowerSchool. (2023). *Schoology learning management system*.
<https://www.powerschool.com>

Sun, P. C., Tsai, R. J., Finger, G., Chen, Y. Y., & Yeh, D. (2008). What drives a successful e-Learning? An empirical investigation of the critical factors influencing learner satisfaction. *Computers & Education*, 50(4), 1183-1202.
<https://doi.org/10.1016/j.compedu.2006.11.007>

Technological University of the Philippines. (2023). *Learning management system implementation*. <https://www.tup.edu.ph>

University of the Philippines. (2023). *University Virtual Learning Environment (UVLe)*. <https://uvle.up.edu.ph>

Batangas State University. (2023). *Smart campus initiative*.
<https://batstate-u.edu.ph>

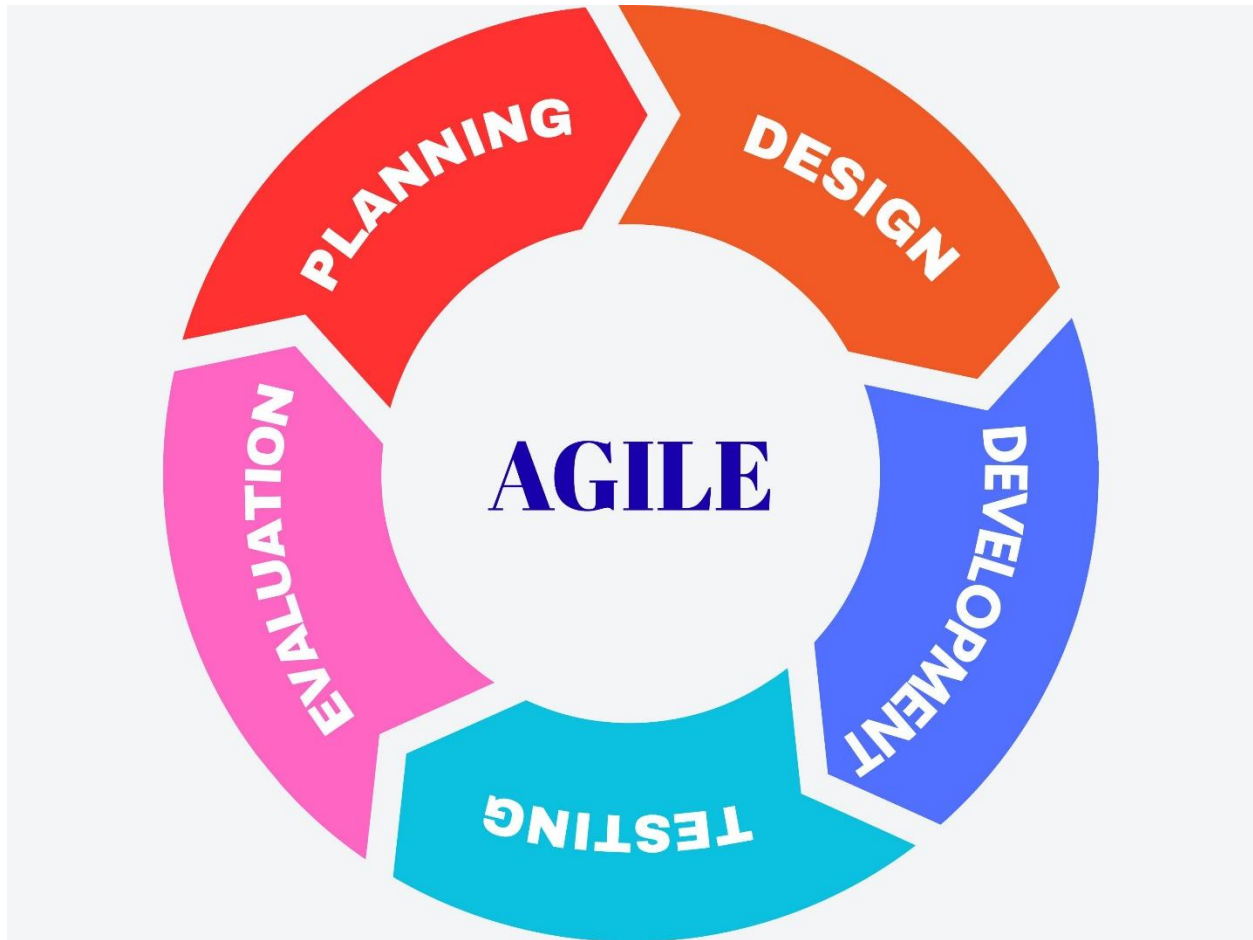
Atlas Curriculum Mapping. (2023). *Atlas curriculum management system*.
<https://www.onatlas.com>

CHAPTER III

METHODOLOGY

The descriptive developmental approach was employed in this study. This approach involves the systematic process of analyzing existing problems, designing a solution, developing the proposed system, and evaluating its effectiveness. It focuses on the creation and assessment of an educational system to ensure that it meets specific instructional and technical standards. In this study, the approach was used to develop a Blended Learning System with Syllabus Mapping that addresses issues in lesson tracking, curriculum alignment, and monitoring of instructional progress.

System Design



In the developmental phase of this study, the Agile Model will be adopted as a guiding framework in designing the proposed syllabus mapping structure. The Agile Model emphasizes flexibility, continuous improvement, and iterative development, which are essential in refining educational systems. Since the proposed syllabus mapping framework will be developed based on feedback from faculty members and students, the Agile approach allows revisions and improvements to be made progressively. This ensures that the developed framework remains responsive to actual instructional needs and adaptable to changes within the blended learning environment

Phases of Agile Model

Planning

During this phase, we began by identifying the primary problems encountered in the current blended learning setup of **I-tech College Inc.** We conducted informal interviews with faculty members and students to understand how syllabi were being implemented and monitored. Based on these discussions, we found that although LMS platforms were being used for uploading materials and submitting assignments, the structured mapping between course outcomes, weekly lessons, and assessment tasks was not systematically integrated into the system. From these findings, we established the main objectives of the project is to develop a blended learning system with an integrated syllabus mapping feature, to align course learning outcomes with lessons, activities, and assessments and to provide monitoring tools for instructors and administrators to track outcome attainment. We also identified system requirements such as user authentication, course creation modules, syllabus encoding, outcome-tagging features, grading system integration, and reporting dashboards for academic monitoring.

Design

After gathering the requirements, we proceeded to design the system architecture and interface. The system was structured into major modules: user management (admin, instructor, and student), course management, syllabus mapping module, assessment module, and analytics dashboard. The syllabus mapping feature was carefully designed to allow instructors to input Intended Learning Outcomes (ILOs), align them with weekly topics, and tag specific activities and assessments to each outcome. We created flowcharts and entity-relationship diagrams to illustrate how syllabus components would be linked within the database. The user interface was designed to be simple and intuitive to ensure ease of navigation for both instructors and students. Wireframes were developed to visualize how instructors could map lessons to outcomes and how administrators could monitor alignment through summary reports.

Development

In this phase, we implemented the core functionalities of the system using an iterative Agile approach. Development was divided into short

sprints, with each sprint focusing on a specific module. The first sprint focused on user authentication and role-based access control. The second sprint implemented course creation and content upload features to support blended learning delivery. The third sprint concentrated on the syllabus mapping module, where instructors could encode course outcomes and connect them to lesson plans and assessments. We ensured that the database structure supported relational mapping between outcomes, instructional materials, and evaluation tools. The grading module was integrated with the outcome-tagging system so that student performance could be analyzed based on specific learning outcomes.

Testing

After completing a functional version of the system, we conducted iterative testing to evaluate performance, usability, and alignment accuracy. We performed unit testing for individual modules and integration testing to ensure that all components worked as a unified system. Particular attention was given to the syllabus mapping feature to verify that outcomes were correctly linked to lessons and assessments. We tested scenarios where instructors edited outcomes, modified lesson plans, or updated assessment criteria to ensure that the changes were reflected consistently throughout the system. User acceptance testing was conducted with selected instructors and students. Feedback was gathered regarding system usability, clarity of the mapping interface, and overall functionality. Identified errors and usability concerns were addressed in subsequent iterations.

Evaluation

Even during early deployment, we continued to gather feedback from users to assess system effectiveness. Evaluation focused on measuring how well the system improved syllabus monitoring, constructive alignment, and blended learning delivery. We assessed whether instructors were able to clearly align learning outcomes with instructional activities and whether administrators could generate reports on outcome attainment. Observations and user feedback served as the basis for further refinements.

The Agile approach allowed us to continuously enhance the system by incorporating suggestions and addressing technical issues in successive development cycles. This iterative refinement ensured that the blended learning system with syllabus mapping evolved according to user needs and institutional requirements.

Research Design

The Blended Learning system with Syllabus Mapping adopts a descriptive developmental approach. The research team will be in contact with I-Tech College's administrator to gain information. The descriptive aspect focuses on analyzing existing practices, particularly in terms of curriculum alignment, instructional quality, faculty readiness, and student engagement. At the same time, the developmental aspect aims to use the findings as a basis for designing an improved syllabus mapping framework that can strengthen the coherence between learning outcomes, instructional strategies, and assessment methods. This design is appropriate because it allows the study to both assess the present situation and provide a concrete, research-based output that contributes to the enhancement of blended learning implementation.

Locale of the Study

The study will be conducted in **I-Tech College Inc. Bago City, Negros Occidental**. The institution utilizes a Learning Management System (LMS) to support online instruction alongside traditional face-to-face classes. The locale is selected due to its active implementation of blended learning, making it relevant to the objectives of the study.

Sample Size

The sample size consists of instructors, administrator and student involved in the blended learning system.

Participant Population	Number of Respondents
Administrator	1
Instructor	8
Student	34

Sampling Technique

Data collection from faculty members and students will utilize convenience sampling. The respondents will be selected based on their

availability and active involvement in the blended learning system of the institution. Since both faculty and students are accessible within the selected departments, this method allows efficient and practical data collection within a limited timeframe. This approach ensures a smooth and manageable gathering of relevant information necessary for the study.

Respondents of the Study

The primary respondents of this study are the instructor and students of I-Tech College Inc. Instructors will provide information regarding course design, syllabus mapping practices, and instructional alignment, while the students provide the feedback on engagement, clarity of learning outcomes, and alignment of activities and assessments. Their responses are essential in describing current practices and identifying areas for development.

Research Instrument

The researchers will use a researcher-made questionnaire as the primary instrument for data collection.

Data Gathering Procedure

1. Observation - The research team will observe the current instructional strategies in I-Tech College Inc. to examine their way

of teaching. This gives us a perspective on how to arrange our system to make it easier for students and instructors.

2. Interview - The researcher will conduct interviews with selected faculty members to better understand their experiences, challenges, and current practices in implementing syllabus mapping within the blended learning system. Through these interviews, the researcher aims to identify specific concerns and gather suggestions that will serve as a basis in improving and developing a more structured syllabus mapping framework.

3. Survey Questionnaire - The researcher will distribute survey questionnaires to both faculty members and students to collect quantitative data regarding the effectiveness of the current blended learning system and syllabus mapping practices. The responses gathered will be analyzed and used as a foundation in designing and enhancing the proposed blended learning system with improved syllabus mapping alignment.

Data Analysis Procedure

The data gathered from the respondents were systematically organized, tabulated, and analyzed using descriptive statistical methods. The primary statistical tool used in this study was the weighted mean, which determined the overall level of acceptability of the proposed Blended Learning System with Syllabus Mapping in terms of functionality, usability, reliability, and efficiency.

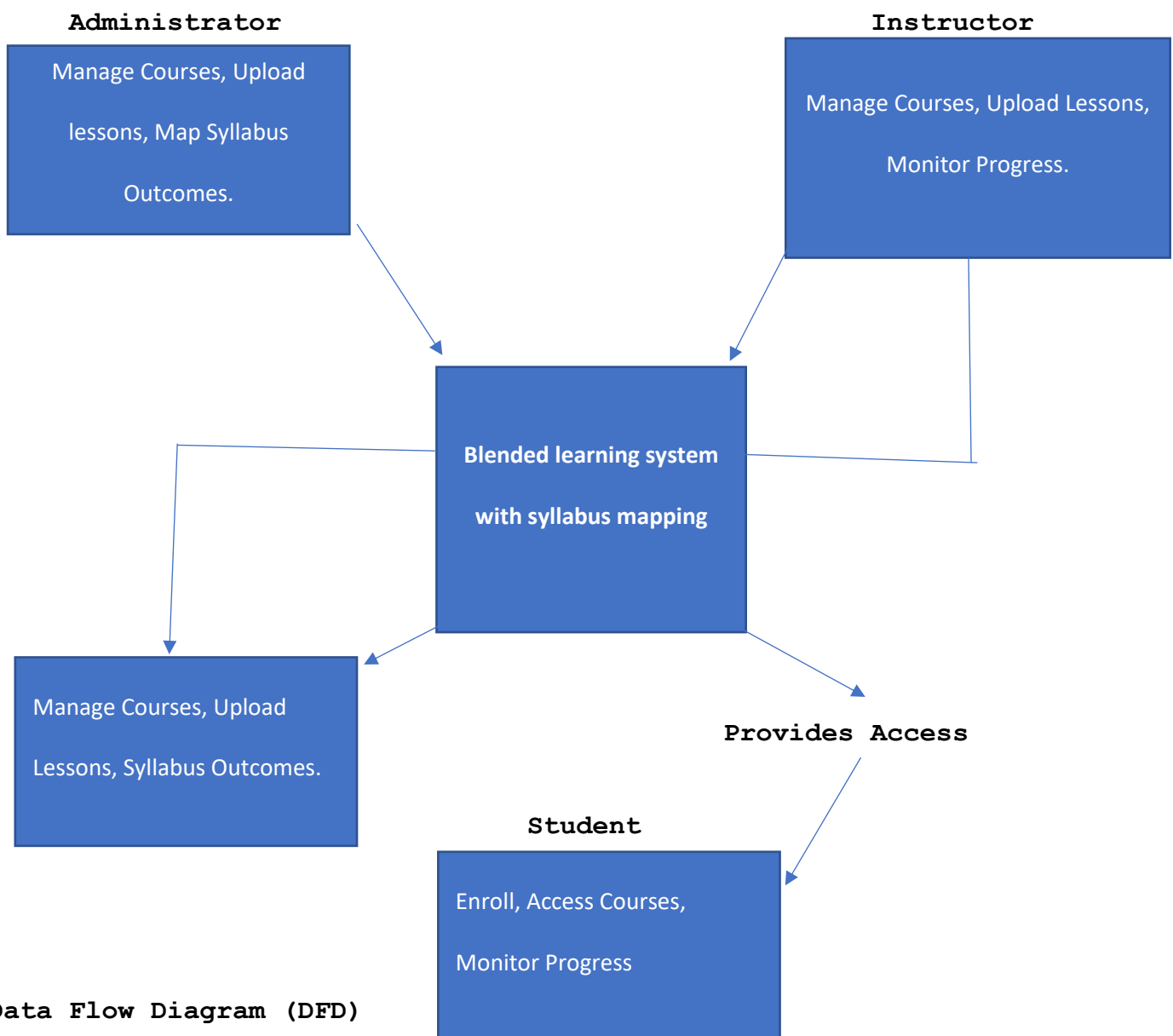
The weighted mean was computed to summarize the respondents' ratings based on the five-point Likert scale. This method allowed the researchers to determine the general perception of the users regarding the system's performance and effectiveness.

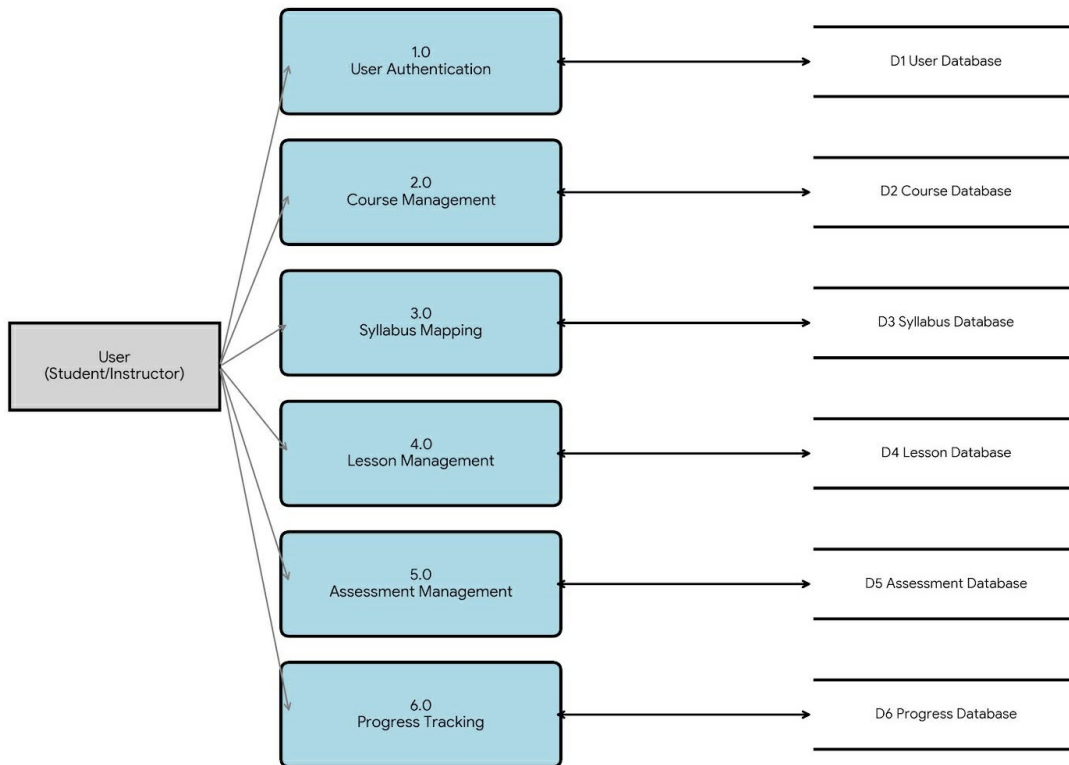
Context Flow Diagram

The Context Flow Diagram illustrates the interaction between external entities and the system. The main external entities include:

- Administrator
- Instructor
- Student

These users interact with the Blended Learning System to manage courses, upload lessons, map syllabus outcomes, enroll students, and monitor progress.





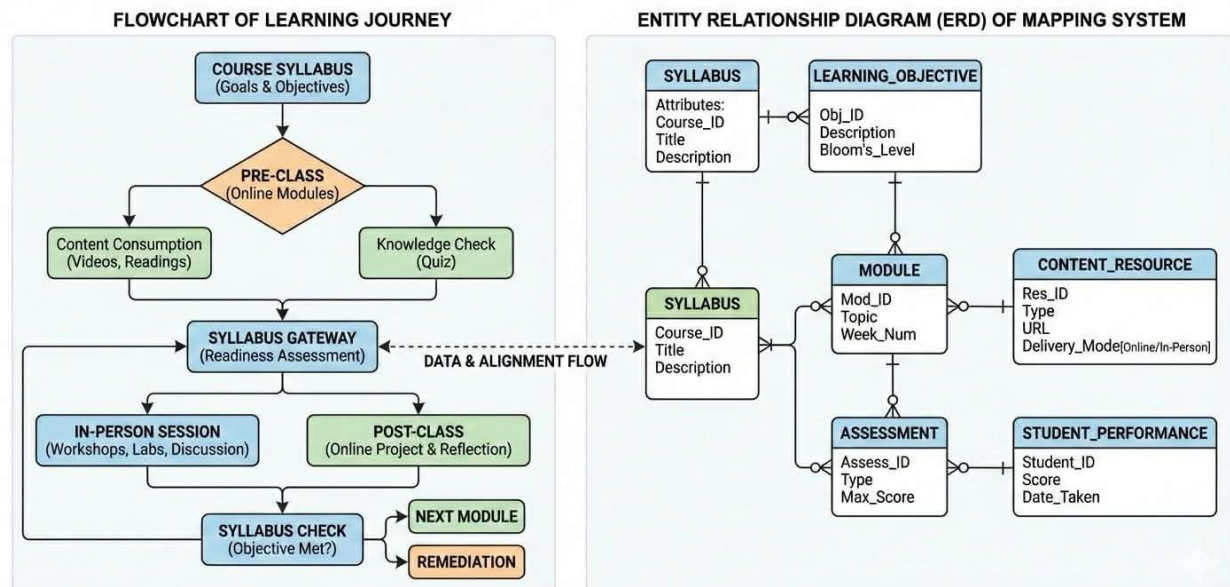
The Data Flow Diagram shows the internal processes of the system, including:

- User Authentication Process
- Course Management Process
- Syllabus Mapping Process
- Lesson Management Process
- Assessment Management Process
- Progress Tracking Process

The system stores data in the following data stores:

- User Database
- Course Database
- Syllabus Database
- Lesson Database
- Assessment Database
- Progress Database

Entity-Relationship Diagram (ERD)

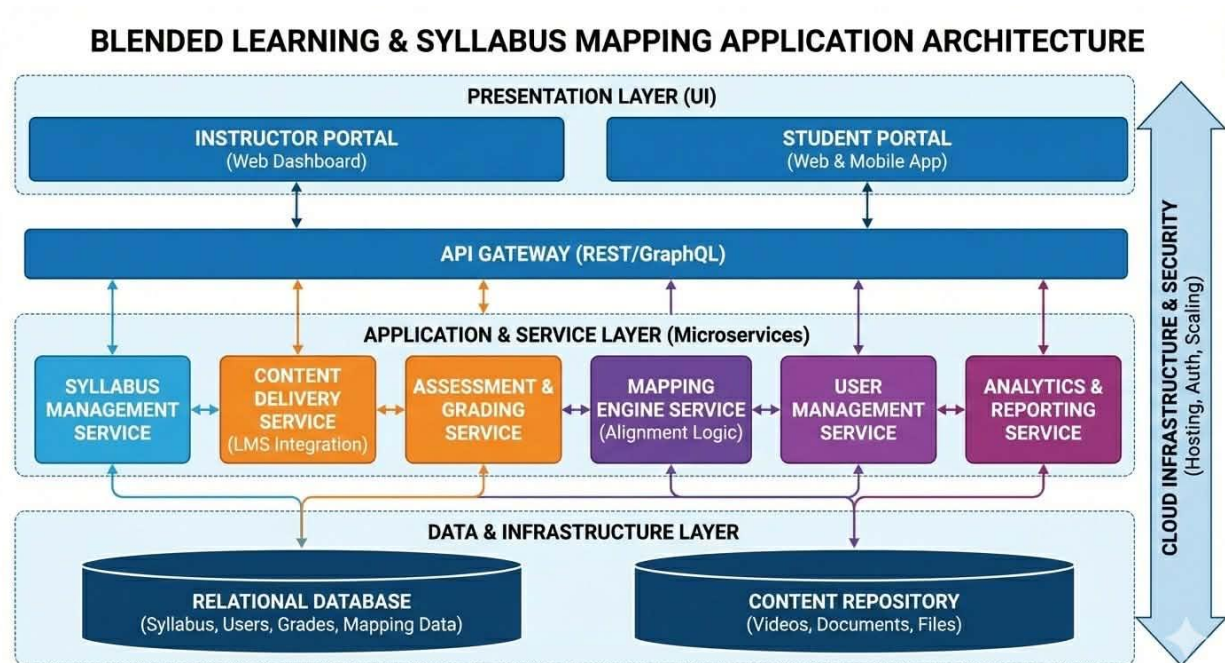


The ERD represents the database structure of the system. The main entities include:

- User
- Student
- Instructor
- Course
- Syllabus
- Learning Outcome
- Lesson
- Assessment
- Progress Tracking

These entities are connected through primary and foreign key relationships to ensure structured data management and accurate tracking of syllabus alignment.

Application Architecture

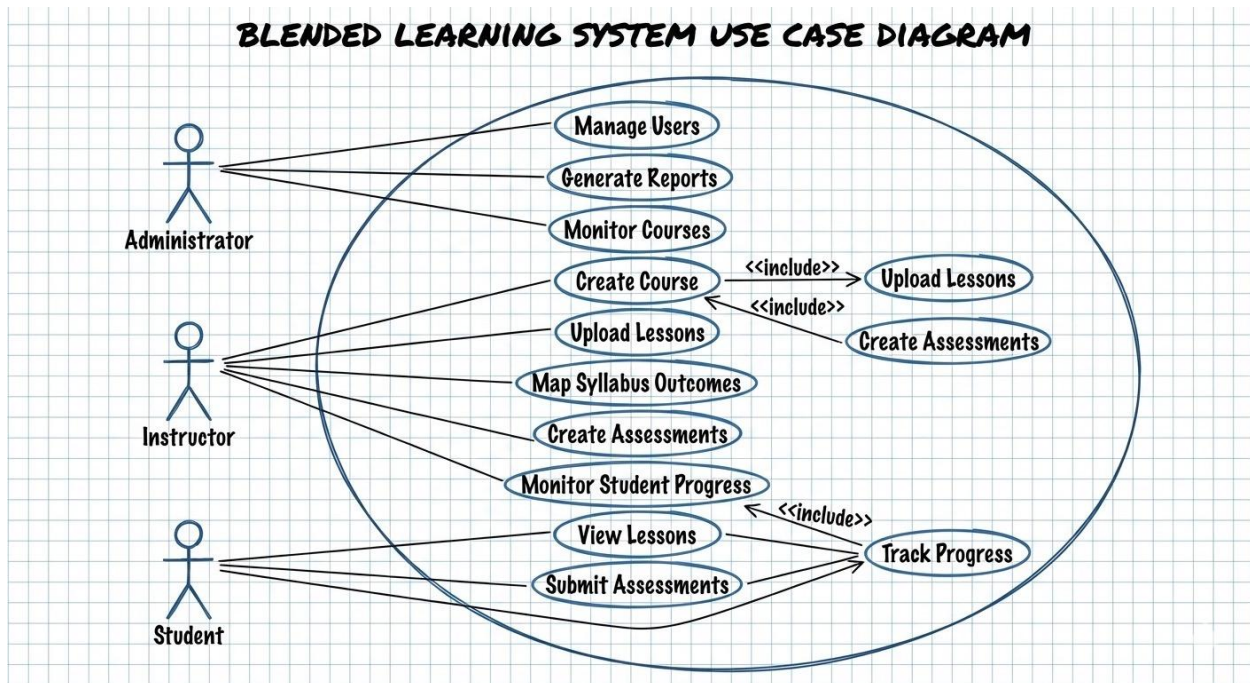


The system follows a three-tier architecture:

1. Presentation Layer - User Interface
2. Application Layer - System Logic
3. Database Layer - Data Storage

This architecture ensures system scalability, security, and maintainability.

Use Case Diagram



The Use Case Diagram includes the following actors:

Administrator:

- Manage Users
- Generate Reports
- Monitor Courses

Instructor:

- Create Course
- Upload Lessons
- Map Syllabus Outcomes
- Create Assessments
- Monitor Student Progress

Student:

- View Lessons
- Submit Assessments
- Track Progress

User Interface Design

The system interface was designed to be simple, user-friendly, and responsive. The dashboard displays course progress, syllabus alignment status, and lesson tracking summaries. Navigation menus allow easy access to modules such as Courses, Lessons, Syllabus Mapping, Assessments, and Reports.

Software Requirements

The system requires the following software:

- Web Browser (Google Chrome, Microsoft Edge)
- Web Server (Apache or equivalent)
- Programming Language (PHP / JavaScript / etc.)
- Database Management System (MySQL or equivalent)

Hardware and Other Required Devices

- Desktop Computer or Laptop
- Minimum 4GB RAM
- Stable Internet Connection
- Mobile device (optional for students)

